

Exploring the BRFSS data

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Setup

Load packages

```
library(ggplot2)
library(dplyr)
```

Load data

```
load("brfss2013.RData")
```

Dimensions of the dataset

```
dim(brfss2013)
```

```
## [1] 491775    330
```

There are 491,775 cases and 330 vectors.

Part 1: Data

The Behavioral Risk Factor Surveillance System (BRFSS) is a collaborative effort between the CDC and all 50 states, District of Columbia, and other U.S. territories to gather data on individual health related issues through telephone survey. The goal of BRFSS is to gather data by state concerning preventative health practices and risk behaviors linked to chronic diseases, injuries, and preventable infectious diseases. Telephone surveys are done through landline and cellular line.

For landline surveys, a random adult was chosen per household. For cellular telephone, individuals who have private residence or college housing were chosen. Both forms of survey deployments involved the non-institutionalized population. Optional questions were included beginning in 2013 involving childhood health for those 17 or younger.

Scope of Inference: Generalizability and Causality Only those who were non-institutionalized were able to take the survey, therefore the data reflects a more narrow population than otherwise. The population that does not have access to landlines or cellphones are not represented. Finally, the 491,775 surveyed may only reflect those that are willing and/or have the time to take the survey.

BRFSS resembles a cross-sectional survey, therefore only inferences of correlation may be made, not causality. Because of the way the survey is deployed, there may exist recall bias, e.g. asking for health information regarding the past 30 days. There may also exist proxy bias, as adults answer for children. Finally, self-reporting bias can exist as information cannot be verified.

Part 2: Research questions

Research question 1: Are there differences in the severity of diabetes across ethnic groups? Are there, also, differences between sex? This question is of interest, because it may reveal groups that are at risk of advanced diabetes.

Research question 2: Is there an association between the days reported of mental health not good and the number of hours slept? The question is of interest because if an association is found, further investigation into causality could improve treatments for sleep or mental health.

Research question 3: In 2013, is there disparity in the income bracket of \$75,000 or more between sex? This question is important to ask because it may reveal efficacy of equality policies at all levels concerning the workplace.

Part 3: Exploratory data analysis

Research question 1:

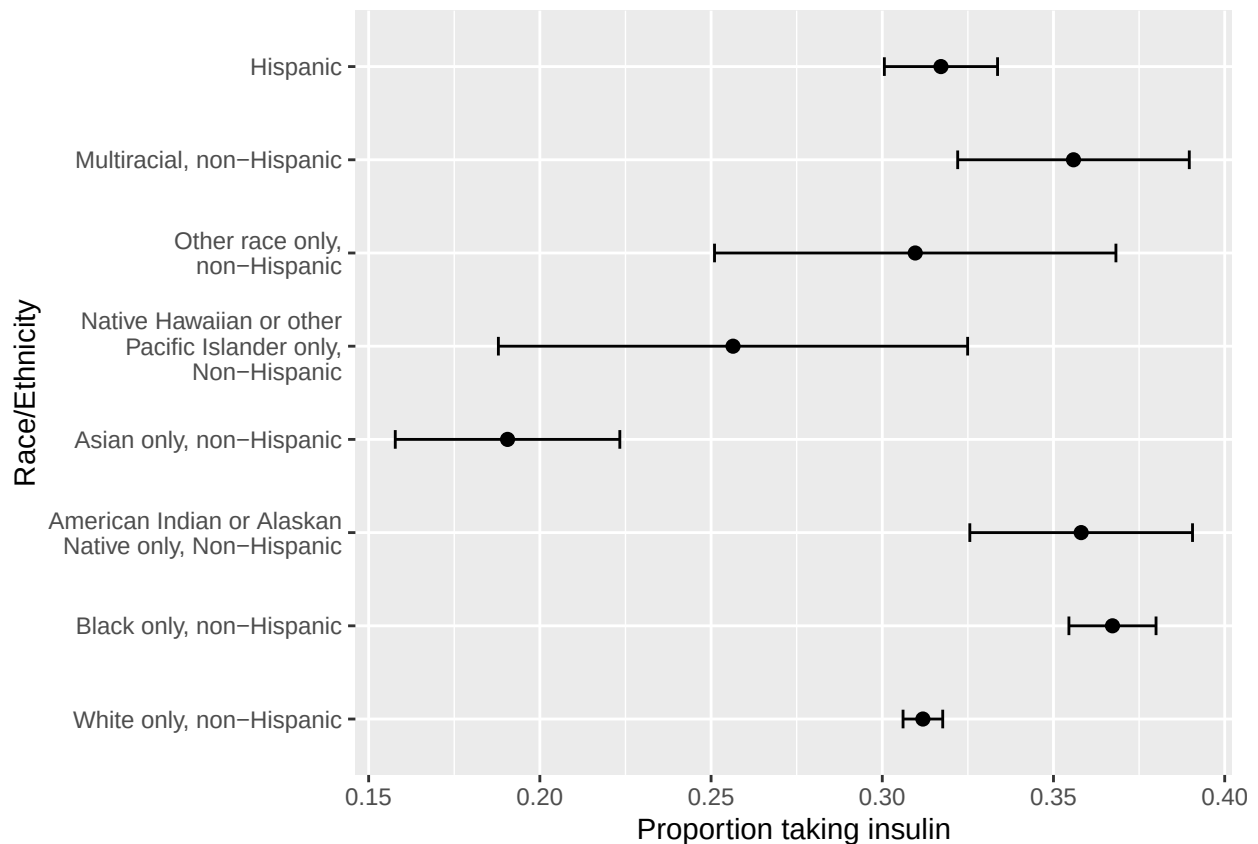
```
d_prop_summary <- brfss2013 %>%
  filter(!is.na(X_race), diabete3 == "Yes",
  insulin %in% c("Yes", "No")) %>%
  group_by(X_race) %>%
  summarize(
    n = n(),
    p_hat = mean(insulin == "Yes"),
    se = sqrt(p_hat * (1 - p_hat) / n),
    lower = p_hat - 1.96 * se,
    upper = p_hat + 1.96 * se
  ) %>%
  arrange(desc(p_hat))
d_prop_summary
```

```
## # A tibble: 8 x 6
##   X_race                n p_hat      se lower upper
##   <fct>                <int> <dbl> <dbl> <dbl> <dbl>
## 1 Black only, non-Hispanic    5517 0.367 0.00649 0.355 0.380
## 2 American Indian or Alaskan Native only, Non-H~    835 0.358 0.0166 0.326 0.391
## 3 Multiracial, non-Hispanic    770 0.356 0.0173 0.322 0.390
## 4 Hispanic                 3046 0.317 0.00843 0.301 0.334
```

## 5 White only, non-Hispanic	24559	0.312	0.00296	0.306	0.318
## 6 Other race only, non-Hispanic	239	0.310	0.0299	0.251	0.368
## 7 Native Hawaiian or other Pacific Islander onl~	156	0.256	0.0350	0.188	0.325
## 8 Asian only, non-Hispanic	551	0.191	0.0167	0.158	0.223

The above summary table shows proportion of insulin takers with diabetes, in descending order, grouped by ethnicity.

```
ggplot(d_prop_summary,
  aes(x = p_hat, y = X_race)) +
  geom_point(size = 2) +
  geom_errorbar(aes(xmin = lower, xmax = upper),
    width = 0.2) +
  scale_y_discrete(labels = function(x) stringr::str_wrap(x, width = 28)) +
  labs(
    x = "Proportion taking insulin",
    y = "Race/Ethnicity",
  )
```



The above graph displays a point plot with 95% confidence intervals based on proportion of insulin takers grouped by ethnicity.

```
d_prop_summary_sex <- brfss2013 %>%
  filter(!is.na(X_race), diabete3 == "Yes",
  insulin %in% c("Yes", "No")) %>%
  group_by(X_race, sex) %>%
  summarize(
    n = n(),
    p_hat = mean(insulin == "Yes"),
    se = sqrt(p_hat * (1 - p_hat) / n),
    lower = p_hat - 1.96 * se,
    upper = p_hat + 1.96 * se
  )
```

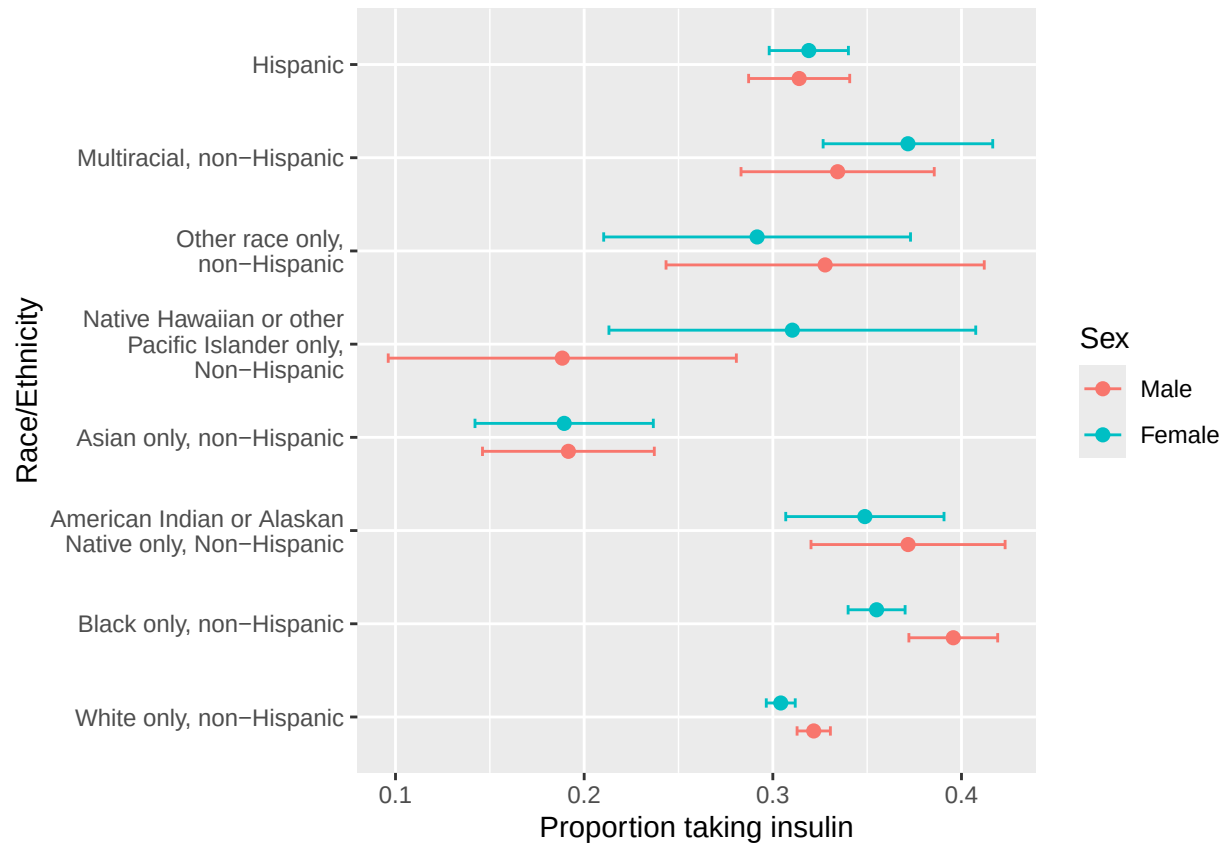
'summarise()' has grouped output by 'X_race'. You can override using the
'.groups' argument.

```
d_prop_summary_sex
```

```
## # A tibble: 16 x 7
## # Groups:   X_race [8]
##   X_race                sex      n p_hat      se lower upper
##   <fct>                <fct> <int> <dbl>   <dbl> <dbl> <dbl>
## 1 White only, non-Hispanic Male  10756 0.322 0.00450 0.313 0.331
## 2 White only, non-Hispanic Fema~ 13803 0.304 0.00392 0.297 0.312
## 3 Black only, non-Hispanic Male   1658 0.396 0.0120 0.372 0.419
## 4 Black only, non-Hispanic Fema~  3859 0.355 0.00770 0.340 0.370
## 5 American Indian or Alaskan Native onl~ Male    339 0.372 0.0262 0.320 0.423
## 6 American Indian or Alaskan Native onl~ Fema~   496 0.349 0.0214 0.307 0.391
## 7 Asian only, non-Hispanic Male    287 0.192 0.0232 0.146 0.237
## 8 Asian only, non-Hispanic Fema~   264 0.189 0.0241 0.142 0.237
## 9 Native Hawaiian or other Pacific Isla~ Male    69 0.188 0.0471 0.0961 0.281
## 10 Native Hawaiian or other Pacific Isla~ Fema~   87 0.310 0.0496 0.213 0.408
## 11 Other race only, non-Hispanic Male   119 0.328 0.0430 0.243 0.412
## 12 Other race only, non-Hispanic Fema~  120 0.292 0.0415 0.210 0.373
## 13 Multiracial, non-Hispanic Male    326 0.334 0.0261 0.283 0.386
## 14 Multiracial, non-Hispanic Fema~   444 0.372 0.0229 0.327 0.417
## 15 Hispanic Male   1153 0.314 0.0137 0.287 0.341
## 16 Hispanic Fema~  1893 0.319 0.0107 0.298 0.340
```

The above summary table shows porportion of insulin takers with diabetes grouped by ethnicity and sex.

```
ggplot(d_prop_summary_sex,
  aes(y = X_race, x = p_hat, color = sex, group = sex)) +
  geom_point(position = position_dodge(width = 0.6),
    size = 2) +
  geom_errorbar(aes(xmin = lower, xmax = upper),
    position = position_dodge(width = 0.6),
    width = 0.2) +
  scale_y_discrete(labels = function(x) stringr::str_wrap(x, width = 28)) +
  labs(
    y = "Race/Ethnicity",
    x = "Proportion taking insulin",
    color = "Sex"
  )
```



The above graph displays proportion of insulin takers grouped by race and sex along with confidence intervals.

Research question 2:

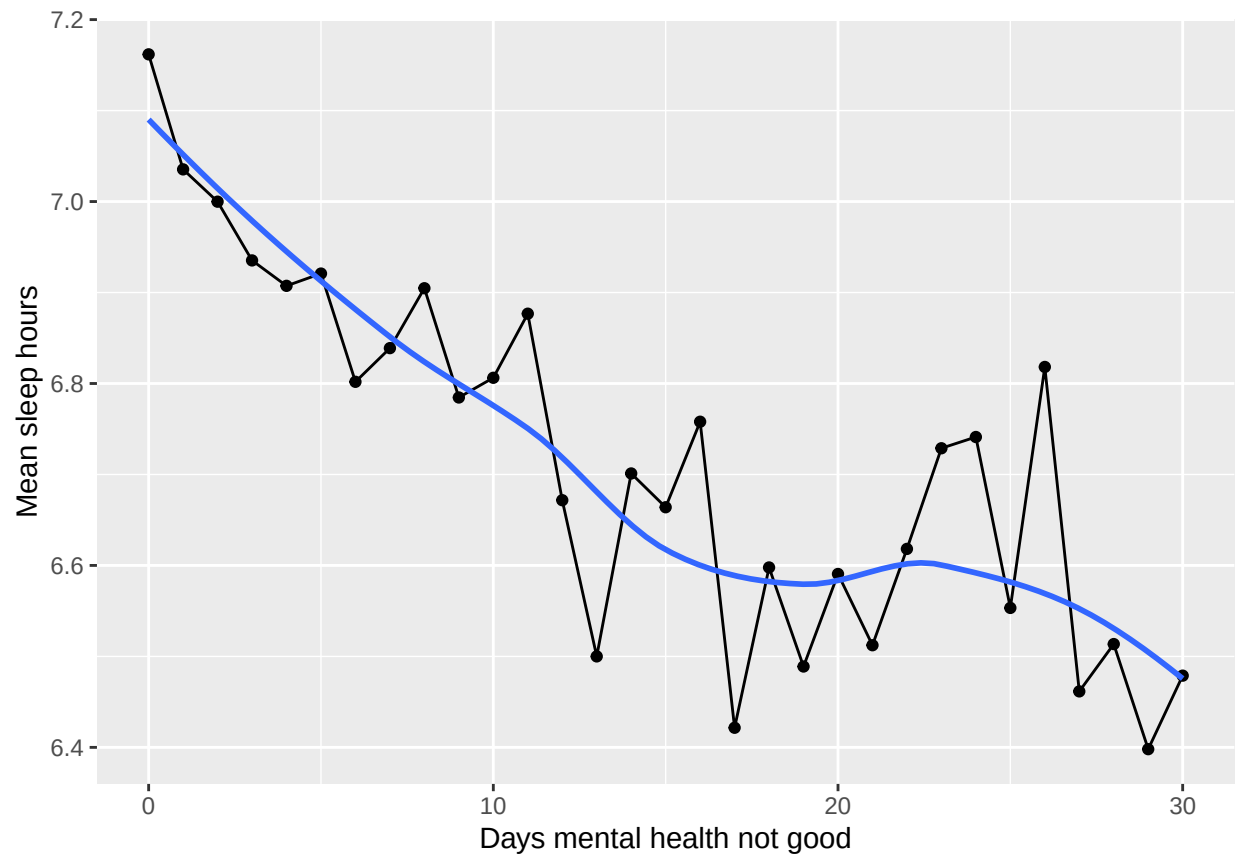
```
mhealth_sleep <- brfss2013 %>%
  filter(!is.na(menthlth),
         !is.na(sleptim1),
         menthlth <= 30,
         sleptim1 > 0) %>%
  group_by(menthlth) %>%
  summarize(
    mean_sleep = mean(sleptim1),
    n = n(),
  )
print(mhealth_sleep, n = Inf)
```

```
## # A tibble: 31 x 3
##   menthlth mean_sleep     n
##   <int>     <dbl> <int>
## 1      0      7.16 329984
## 2      1      7.04 15124
## 3      2      7.00 23343
## 4      3      6.94 13493
## 5      4      6.91  6594
## 6      5      6.92 16549
```

##	7	6	6.80	1842
##	8	7	6.84	6293
##	9	8	6.90	1228
##	10	9	6.78	195
##	11	10	6.81	11774
##	12	11	6.88	73
##	13	12	6.67	795
##	14	13	6.5	94
##	15	14	6.70	2469
##	16	15	6.66	10746
##	17	16	6.76	157
##	18	17	6.42	102
##	19	18	6.60	179
##	20	19	6.49	45
##	21	20	6.59	6544
##	22	21	6.51	488
##	23	22	6.62	110
##	24	23	6.73	59
##	25	24	6.74	85
##	26	25	6.55	2279
##	27	26	6.82	88
##	28	27	6.46	156
##	29	28	6.51	664
##	30	29	6.40	407
##	31	30	6.48	24655

```
ggplot(mhealth_sleep, aes(x = menthlth, y = mean_sleep)) +
  geom_point() +
  geom_line() +
  geom_smooth(se = FALSE) +
  labs(x = "Days mental health not good", y = "Mean sleep hours")
```

```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



Research question 3:

```
income_sex <- brfss2013 %>%
  filter(!is.na(sex),
         !is.na(income2),
         ) %>%
  group_by(X_state, sex) %>%
  summarize(phat = mean(income2 == "$75,000 or more"), .groups = "drop")
print(income_sex, n = Inf)
```

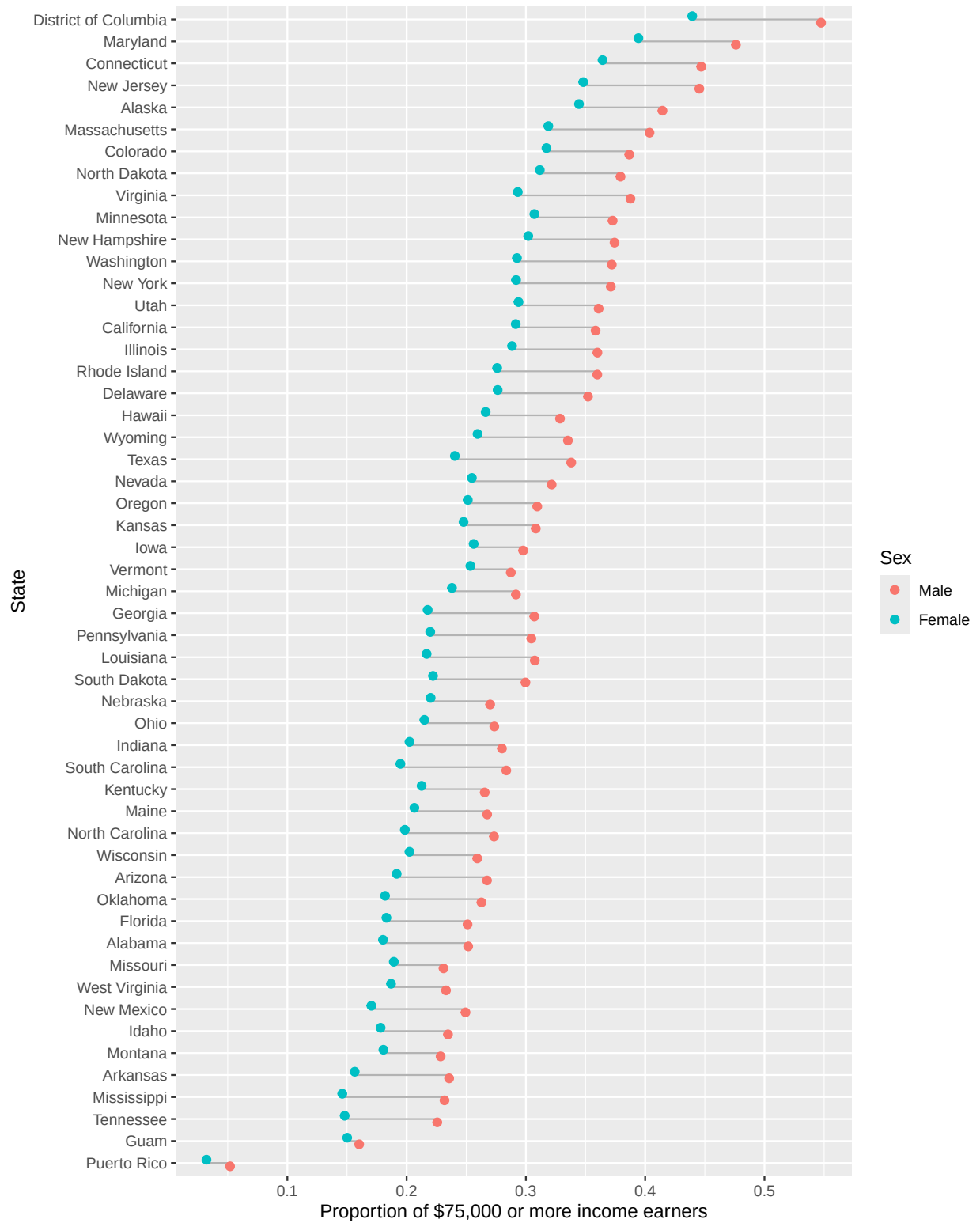
```
## # A tibble: 106 x 3
##   X_state      sex    phat
##   <fct>      <fct> <dbl>
## 1 Alabama   Male    0.252
## 2 Alabama   Female  0.180
## 3 Alaska    Male    0.415
## 4 Alaska    Female  0.345
## 5 Arizona   Male    0.267
## 6 Arizona   Female  0.192
## 7 Arkansas  Male    0.236
## 8 Arkansas  Female  0.156
## 9 California Male    0.358
## 10 California Female  0.291
## 11 Colorado  Male    0.387
## 12 Colorado  Female  0.317
## 13 Connecticut Male    0.447
```

##	14 Connecticut	Female	0.364
##	15 Delaware	Male	0.352
##	16 Delaware	Female	0.276
##	17 District of Columbia	Male	0.547
##	18 District of Columbia	Female	0.439
##	19 Florida	Male	0.251
##	20 Florida	Female	0.183
##	21 Georgia	Male	0.307
##	22 Georgia	Female	0.218
##	23 Hawaii	Male	0.329
##	24 Hawaii	Female	0.266
##	25 Idaho	Male	0.235
##	26 Idaho	Female	0.178
##	27 Illinois	Male	0.360
##	28 Illinois	Female	0.288
##	29 Indiana	Male	0.280
##	30 Indiana	Female	0.202
##	31 Iowa	Male	0.298
##	32 Iowa	Female	0.256
##	33 Kansas	Male	0.308
##	34 Kansas	Female	0.248
##	35 Kentucky	Male	0.265
##	36 Kentucky	Female	0.213
##	37 Louisiana	Male	0.307
##	38 Louisiana	Female	0.217
##	39 Maine	Male	0.268
##	40 Maine	Female	0.207
##	41 Maryland	Male	0.476
##	42 Maryland	Female	0.394
##	43 Massachusetts	Male	0.404
##	44 Massachusetts	Female	0.319
##	45 Michigan	Male	0.292
##	46 Michigan	Female	0.238
##	47 Minnesota	Male	0.373
##	48 Minnesota	Female	0.307
##	49 Mississippi	Male	0.232
##	50 Mississippi	Female	0.146
##	51 Missouri	Male	0.231
##	52 Missouri	Female	0.189
##	53 Montana	Male	0.229
##	54 Montana	Female	0.181
##	55 Nebraska	Male	0.270
##	56 Nebraska	Female	0.220
##	57 Nevada	Male	0.322
##	58 Nevada	Female	0.255
##	59 New Hampshire	Male	0.374
##	60 New Hampshire	Female	0.302
##	61 New Jersey	Male	0.445
##	62 New Jersey	Female	0.348
##	63 New Mexico	Male	0.249
##	64 New Mexico	Female	0.170
##	65 New York	Male	0.371
##	66 New York	Female	0.292
##	67 North Carolina	Male	0.273

##	68 North Carolina	Female	0.199
##	69 North Dakota	Male	0.379
##	70 North Dakota	Female	0.312
##	71 Ohio	Male	0.274
##	72 Ohio	Female	0.215
##	73 Oklahoma	Male	0.263
##	74 Oklahoma	Female	0.182
##	75 Oregon	Male	0.310
##	76 Oregon	Female	0.251
##	77 Pennsylvania	Male	0.305
##	78 Pennsylvania	Female	0.220
##	79 Rhode Island	Male	0.360
##	80 Rhode Island	Female	0.276
##	81 South Carolina	Male	0.284
##	82 South Carolina	Female	0.195
##	83 South Dakota	Male	0.300
##	84 South Dakota	Female	0.222
##	85 Tennessee	Male	0.226
##	86 Tennessee	Female	0.148
##	87 Texas	Male	0.338
##	88 Texas	Female	0.240
##	89 Utah	Male	0.361
##	90 Utah	Female	0.294
##	91 Vermont	Male	0.287
##	92 Vermont	Female	0.253
##	93 Virginia	Male	0.388
##	94 Virginia	Female	0.293
##	95 Washington	Male	0.372
##	96 Washington	Female	0.293
##	97 West Virginia	Male	0.233
##	98 West Virginia	Female	0.187
##	99 Wisconsin	Male	0.259
##	100 Wisconsin	Female	0.202
##	101 Wyoming	Male	0.335
##	102 Wyoming	Female	0.259
##	103 Guam	Male	0.160
##	104 Guam	Female	0.150
##	105 Puerto Rico	Male	0.0520
##	106 Puerto Rico	Female	0.0322

The above summary table shows grouping by state and sex and the respective proportion that earns an income of \$75,000 or more.

```
ggplot(income_sex,
       aes(x = phat, y = reorder(X_state, phat), color = sex, group = sex)) +
  geom_line(aes(group = X_state), color = "grey70") + # adds horizontal "track"
  geom_point(position = position_dodge(width = 0.6),
             size = 2) +
labs(x = "Proportion of $75,000 or more income earners",
     y = "State",
     color = "Sex")
```



The above point plot shows grouping by state and sex with the proportion of \$75,000 or greater earners arranged in descending order.